Tabla de derivadas

Propiedades de la derivada	
Función	**Derivada**
$y = k \cdot f, k \equiv cte$	$y' = k \cdot f'$
$y = f \pm g$	$y' = f' \pm g'$
$y = f \cdot g$	$y' = f' \cdot g + f \cdot g'$
$y = \frac{f}{g}$	$y' = \frac{f' \cdot g - f \cdot g'}{g^2}$
$y = f^g$	$y' = g \cdot f^{g-1} \cdot f' + f^g \cdot g'$
$y = g \circ f = g(f(x))$	$y' = g'(f) \cdot f'$

Derivada de funciones compuestas		Derivada de funciones simples	
Función	**Derivada**	**Función**	**Derivada**
$y = k, k \equiv cte$	$y' = 0$	$y = x$	$y' = 1$
$y = [f(x)]^n, n \in \mathbb{R}$	$y' = n \cdot [f(x)]^{n-1} \cdot f'(x)$	$y = x^n, n \in \mathbb{R}$	$y' = n \cdot x^{n-1}$
$y = \sqrt[n]{f(x)}, n \in \mathbb{R}$	$y' = \frac{f'(x)}{n \cdot \sqrt[n]{[f(x)]^{n-1}}}$	$y = \sqrt[n]{x}, n \in \mathbb{R}$	$y' = \frac{1}{n \cdot \sqrt[n]{x^{n-1}}}$
$y = \log_a[f(x)]$	$y' = \frac{f'(x)}{f(x)} \cdot \log_a e$	$y = \log_a x$	$y' = \frac{1}{x} \cdot \log_a e$
$y = \ln[f(x)]$	$y' = \frac{f'(x)}{f(x)}$	$y = \ln x$	$y' = \frac{1}{x}$
$y = a^{f(x)}$	$y' = a^{f(x)} \cdot f'(x) \cdot \ln a$	$y = a^x$	$y' = a^x \cdot \ln a$
$y = e^{f(x)}$	$y' = e^{f(x)} \cdot f'(x)$	$y = e^x$	$y' = e^x$
$y = \text{sen}[f(x)]$	$y' = f'(x) \cdot \cos[f(x)]$	$y = \text{sen}(x)$	$y' = \cos(x)$
$y = \cos[f(x)]$	$y' = -f'(x) \cdot \text{sen}[f(x)]$	$y = \cos(x)$	$y' = -\text{sen}(x)$
$y = \text{tg}[f(x)]$	$y' = \frac{f'(x)}{\cos^2[f(x)]} = f'(x)[1 + \text{tg}^2[f(x)]]$	$y = \text{tg}(x)$	$y' = \frac{1}{\cos^2(x)} = 1 + \text{tg}^2[f(x)]$
$y = \text{cosec}[f(x)]$	$y' = -\text{cosec}[f(x)] \cdot \text{cotg}[f(x)] \cdot f'(x)$	$y = \text{cosec}(x)$	$y' = -\text{cosec}(x) \cdot \text{cotg}(x)$
$y = \sec[f(x)]$	$y' = \sec[f(x)] \cdot \text{tg}[f(x)] \cdot f'(x)$	$y = \sec(x)$	$y' = \sec(x) \cdot \text{tg}(x)$
$y = \text{cotg}[f(x)]$	$y' = -\frac{f'(x)}{\text{sen}^2[f(x)]}$	$y = \text{cotg}(x)$	$y' = -\frac{1}{\text{sen}^2(x)}$
$y = \text{arcsen}[f(x)]$	$y' = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$	$y = \text{arcsen}(x)$	$y' = \frac{1}{\sqrt{1 - x^2}}$
$y = \arccos[f(x)]$	$y' = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$	$y = \arccos(x)$	$y' = -\frac{1}{\sqrt{1 - x^2}}$
$y = \text{arctg}[f(x)]$	$y' = \frac{f'(x)}{1 + [f(x)]^2}$	$y = \text{arctg}(x)$	$y' = \frac{1}{1 + x^2}$
$y = \text{Sh}[f(x)]$	$y' = f'(x) \cdot \text{Ch}[f(x)]$	$y = \text{Sh}(x)$	$y' = \text{Ch}(x)$
$y = \text{Ch}[f(x)]$	$y' = f'(x) \cdot \text{Sh}[f(x)]$	$y = \text{Ch}(x)$	$y' = \text{Sh}(x)$
$y = \text{Th}[f(x)]$	$y' = \frac{f'(x)}{\text{Ch}^2[f(x)]}$	$y = \text{Th}(x)$	$y' = \frac{1}{\text{Ch}^2(x)}$
$y = \text{ArgSh}[f(x)]$	$y' = \frac{f'(x)}{\sqrt{1 + [f(x)]^2}}$	$y = \text{ArgSh}(x)$	$y' = \frac{1}{\sqrt{1 + x^2}}$
$y = \text{ArgCh}[f(x)]$	$y' = \frac{f'(x)}{\sqrt{[f(x)]^2 - 1}}$	$y = \text{ArgCh}(x)$	$y' = \frac{1}{\sqrt{x^2 - 1}}$
$y = \text{ArgTh}[f(x)]$	$y' = \frac{f'(x)}{1 - [f(x)]^2}$	$y = \text{ArgTh}(x)$	$y' = \frac{1}{1 - x^2}$